Finite-Memory Tracking Under Drift

A Cube-Root Law for Useful Memory

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Abstract

Finite-memory tracking under drift faces a basic tension: retaining more past information reduces statistical noise, but it also pushes the estimate further away from the present. As the target distribution moves, each additional sample becomes at once useful and increasingly old, so the problem is not simply how much to remember, but how long remembered evidence should remain operationally current.

We study this trade-off under Wasserstein nonstationarity. For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory and a staleness term that grows with drift. Balancing those two forces yields a **cube-root law**: the **optimal memory horizon** scales as $n^* \asymp (C_K/\zeta)^{2/3}$, while the corresponding **minimum tracking error** scales as $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$. In the critical-window regime, we also prove a minimax lower bound, showing that this finite-memory floor is structural rather than estimator-specific.

The experiments follow the same pattern. Gaussian drift experiments produce the predicted 'U'-curve, sliding-window and EWMA estimators recover the cube-root scaling closely, and dynamic-window illustrations show how the best memory horizon shifts with drift strength over time. A cube-root cap then turns the law into an operational horizon regulator, keeping the effective window near the oracle scale across drift regimes.

The resulting picture is theorem-first and operational: under drift, optimal tracking requires memory that is neither too short nor too old. Viewed through an Age-of-Information lens, the law quantifies how much freshness a finite-memory learner can preserve while tracking a changing world.

Keywords: tracking under drift; finite-memory estimation; Wasserstein distance; Sinkhorn divergence; concept drift; information freshness; Age of Information; adaptive memory.

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1 Introduction

Adaptive systems rarely track a fixed target. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself drifts over time. A learner with finite memory must therefore solve a basic but easy-to-miss problem: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. Using more past data reduces sampling noise, but it also increases the age of the information being used to track the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The central claim of this paper is that this tension induces a quantitative law for finite-memory tracking under Wasserstein nonstationarity.

For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory length and a staleness term that increases with drift and effective data age. Optimizing that balance yields a cube-root law: $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$ with $n^* \asymp (C_K/\zeta)^{2/3}$. The law is exact for the averaging class studied in the paper. It is not asserted as a theorem for every adaptive estimator. What the paper does show beyond that class is a minimax lower bound for window-restricted estimators in the critical-window regime, establishing that the finite-memory floor is structural rather than a quirk of one specific rule.

This viewpoint also gives the paper its interpretive layer. The limiting factor is the age of the information that finite memory forces the estimator to retain, alongside the usual finite-sample cost. The result can therefore be read as a freshness law: the learner pays for variance reduction by carrying older evidence, and optimal tracking occurs at a memory horizon that balances noise against informational aging. In Age-of-Information terms, timeliness belongs to the estimation problem itself.

The empirical section follows the same logic in three steps. Gaussian tracking experiments exhibit the predicted ‘U‘-curve, ablations over drift strength show how the best window shifts with nonstationarity, and dynamic-window illustrations make the operational memory law visible when the local drift rate changes over time. Sinkhorn-based tracking scores serve as the practical measurement layer for this error surface, and the cube-root cap shows how the same law can be used as an online horizon regulator as well as a retrospective explanation.

The scope of the paper is deliberately narrow. We study finite-memory tracking under drift, a lower bound for restricted estimators, and the associated freshness interpretation, together with experiments chosen to validate that story. The paper asks a single question: what tracking accuracy is possible when the learner’s memory is finite and the target distribution keeps moving?

2 Useful Memory Geometry

Under continuous drift, memory is not only a statistical resource but also a temporal liability. The core question is therefore not whether one can react to change, but how long a sample should remain operationally useful before it becomes stale.

Definition 2.1 (Lag-Inclusive Estimation Error)**.** For a finite-memory estimator with effective window size n and drift rate ζ, we write the expected tracking error as

$$\mathcal{E}(n) \;=\; C_K\, n^{-1/2} \;+\tfrac{1}{2}\,\zeta\, n,$$

where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.

The two terms in definition 2.1 pull in opposite directions. Increasing n reduces variance but raises the age of the retained evidence. That tension is the basic geometry of useful memory under drift.

Proposition 2.2 (Optimal Window Under Drift). *Under definition 2.1, the error $\mathcal{E}(n)$ is minimised at*

$$n^* \;=\; \left(\frac{C_K}{\zeta}\right)^{2/3},$$

with minimum value

$$\mathcal{E}_{\min} \;=\; \tfrac{3}{2}\, C_K^{2/3}\, \zeta^{1/3}.$$

Proof. Differentiate $\mathcal{E}(n)$ with respect to n and solve $d\mathcal{E}/dn = 0$. The unique positive critical point gives the stated window. Substituting back yields the minimum value. □

Corollary 2.3 (EWMA Analogue). *For exponential weights $w_j = \alpha(1-\alpha)^j$, the effective lag is $\tau_\alpha = (1-\alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale.*

Definition 2.4 (Effective Memory Age). Let an estimator place weights $w_0, \ldots, w_{n-1}$ on the most recent observations, with $\sum_j w_j = 1$. Its *effective memory age* is

$$A(w) \;=\; \sum_{j=0}^{n-1} j\, w_j,$$

and its corresponding freshness score is the monotone transform $F(w) = 1/(1 + A(w))$. For a uniform window, $A(w) = (n-1)/2$.

The quantity $A(w)$ gives a compact description of the temporal geometry of a finite-memory estimator. Short windows have small age and large variance; long windows have small variance and large age. The cube-root horizon marks the point where the gain from lowering variance no longer compensates for the added age of the retained evidence.

Proposition 2.5 (Finite-Memory Floor). *Let the target trajectory satisfy a W_2-Lipschitz drift bound with rate ζ, and restrict attention to estimators that only use the last m samples. Then, in the critical-window regime, the minimax tracking error over that class cannot beat the same variance-plus-staleness scaling:*

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E}\, d\big(\hat{\mathcal{P}}_m, \mathcal{P}^*\big) \;\gtrsim\; C_K\, m^{-1/2} + \tfrac{1}{2}\,\zeta\, m.$$

Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural floor of finite memory on this drift class.

Proof sketch. The lower bound combines a finite-sample concentration term with a lag term induced by the Lipschitz drift of the oracle trajectory. The concentration part is inherited from debiased Sinkhorn estimation, while the lag term follows from the fact that any estimator restricted to the last m samples must average over observations whose oracle law has moved by order ζm. The two terms are additive at the level of the bound, so the same cube-root balance governs the critical regime. □

Remark 2.6 (Measurement layer). Debiased Sinkhorn divergence serves as a practical measurement layer when the underlying geometry must be estimated from samples. That computational choice supports the tracking law, but the law itself is about the horizon of useful memory, not about any particular OT solver.

3 Empirical Validation

The experiments test a single claim: under continuous drift, memory must be regulated by temporal usefulness rather than by statistical evidence of abrupt change alone. The empirical story has three parts. First, the Gaussian sweep recovers the cube-root law itself. Second, continuous-drift benchmarks show that classical adaptive windows can keep too much stale memory. Third, the piecewise-drift benchmark shows that a cube-root cap can recover the local oracle horizon across regimes.

3.1 Law Validation

We begin with the Gaussian sweep used to validate the finite-memory law. The setup produces the characteristic U-curve: for small windows, error is dominated by variance; for large windows, error is dominated by staleness. The empirical minimum follows the predicted cube-root scaling, and EWMA behaves as the natural finite-memory analogue rather than as a competing theory.

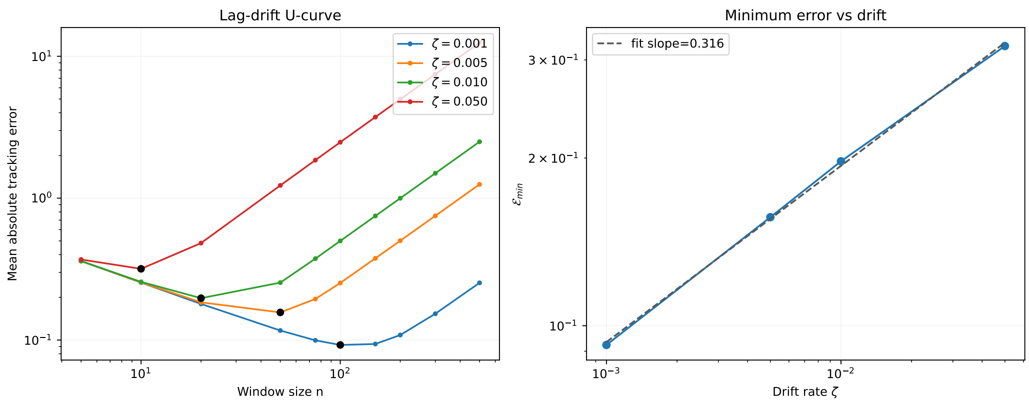

Figure 1: Cube-root validation on the Gaussian sweep. Left: tracking error as a function of window size, showing the predicted U-shape for multiple drift rates. Right: the minimum achievable error follows an approximate $\zeta^{1/3}$ law. This figure validates the structural trade-off between variance and staleness that underlies the rest of the paper.

3.2 Lag–Variance Frontier

The U-curve becomes more informative when it is viewed as a frontier in horizon space. The next figure plots tail error against effective memory horizon for a fixed-window sweep and overlays the operating points of the main methods. The left side is variance-dominated, the right side is stale-memory dominated, and the knee marks the region where useful memory is best preserved. Read through the lens of effective memory age, the same curve says when older evidence stops buying accuracy. The frontier is asymmetric because the variance term decays sublinearly while the staleness term grows linearly. On the left, extra memory still buys a noticeable drop in noise; on the right, extra memory mainly adds age. The knee therefore appears sharper on the stale side than on the variance side.

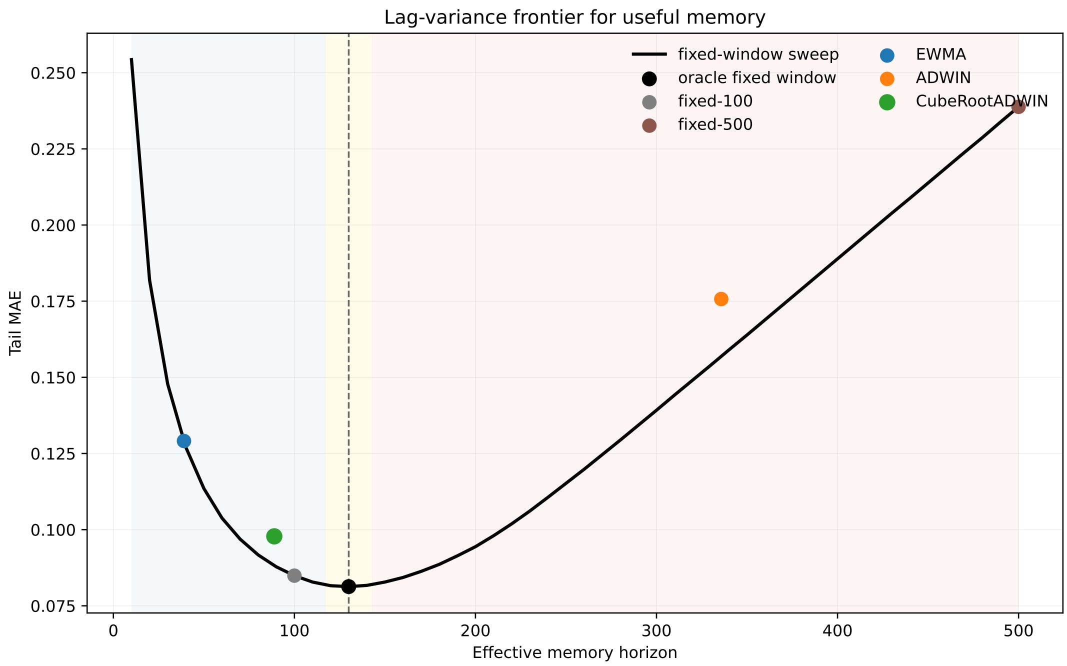

Figure 2: Lag–variance frontier for useful memory. The fixed-window sweep traces the empirical frontier: short horizons are noisy, long horizons are stale, and the knee marks the useful-memory optimum. The frontier is asymmetric because variance decays sublinearly while staleness grows linearly. ADWIN sits far to the stale side, whereas CubeRootADWIN lies close to the frontier knee.

3.3 Over-Memory Under Continuous Drift

The next benchmark focuses on the operational failure mode. We compare five methods on a smooth drifting stream: a short fixed window, a long fixed window, EWMA, ADWIN, and the cube-root capped variant. The comparison asks when memory becomes stale before classical changepoint evidence accumulates.

The important phenomenon here is the separation between cap activation and the Hoeffding test. The cap can activate while the test remains silent, which keeps memory validity distinct from the evidence threshold for abrupt change.

Table 1: Continuous-drift benchmark under a smooth drift rate. The key gap is between statistical evidence and operational freshness: ADWIN keeps a much wider window than the cube-root regulator, while the long fixed window collapses under lag.

Method	Tail MAE	Mean horizon	Drift events	Cap events
Fixed-100	0.0849	100.0	–	–
Fixed-500	0.2388	500.0	–	–
EWMA	0.1291	39.0	–	–
ADWIN	0.1757	335.8	3.9	0.0
CubeRootADWIN	0.0978	88.9	0.0	1374.7

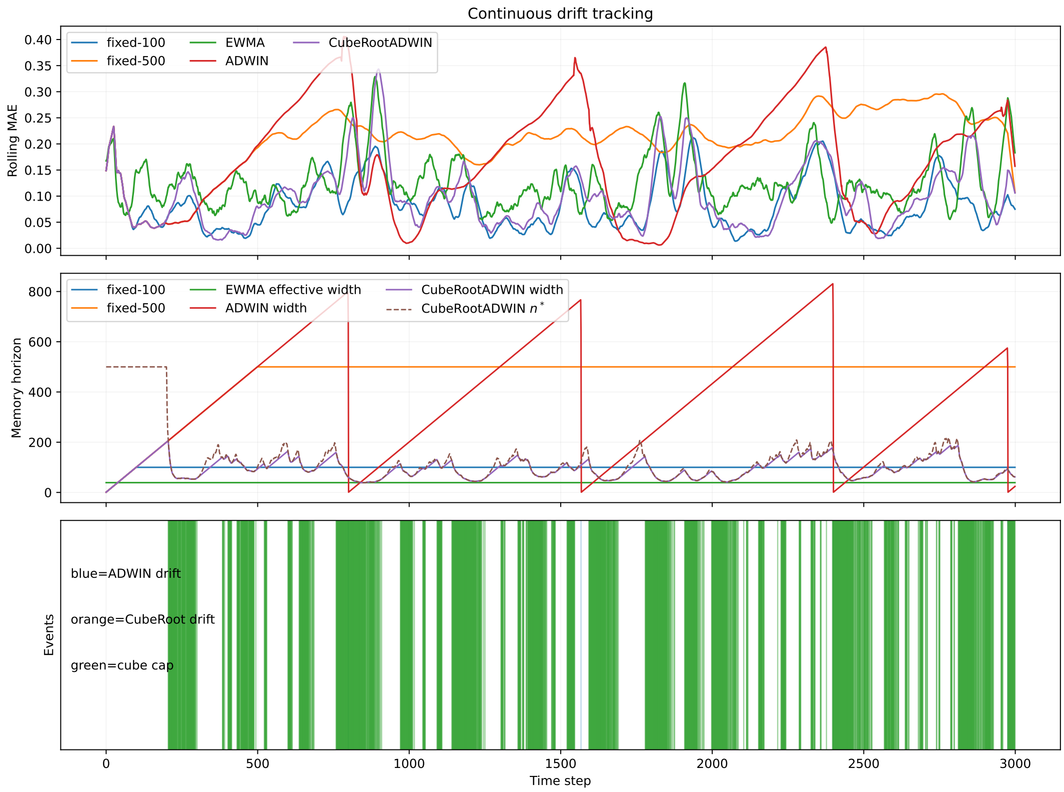

Figure 3: Continuous-drift benchmark. Fixed-500 is the stale-memory baseline; ADWIN retains far more memory than the oracle-scale horizon; CubeRootADWIN pulls the effective horizon toward the useful-memory scale and exposes a ‘cap-only’ regime, where memory becomes operationally stale before statistical changepoint evidence appears.

3.4 Temporal Validity Gap

The gap becomes more explicit when the drift rate varies. The next figure plots the first cap activation of the cube-root regulator and the first changepoint decision of ADWIN as the drift rate changes. In the smooth-drift regime, the cap activates much earlier than the detector; as the drift becomes faster, the gap shrinks but does not disappear.

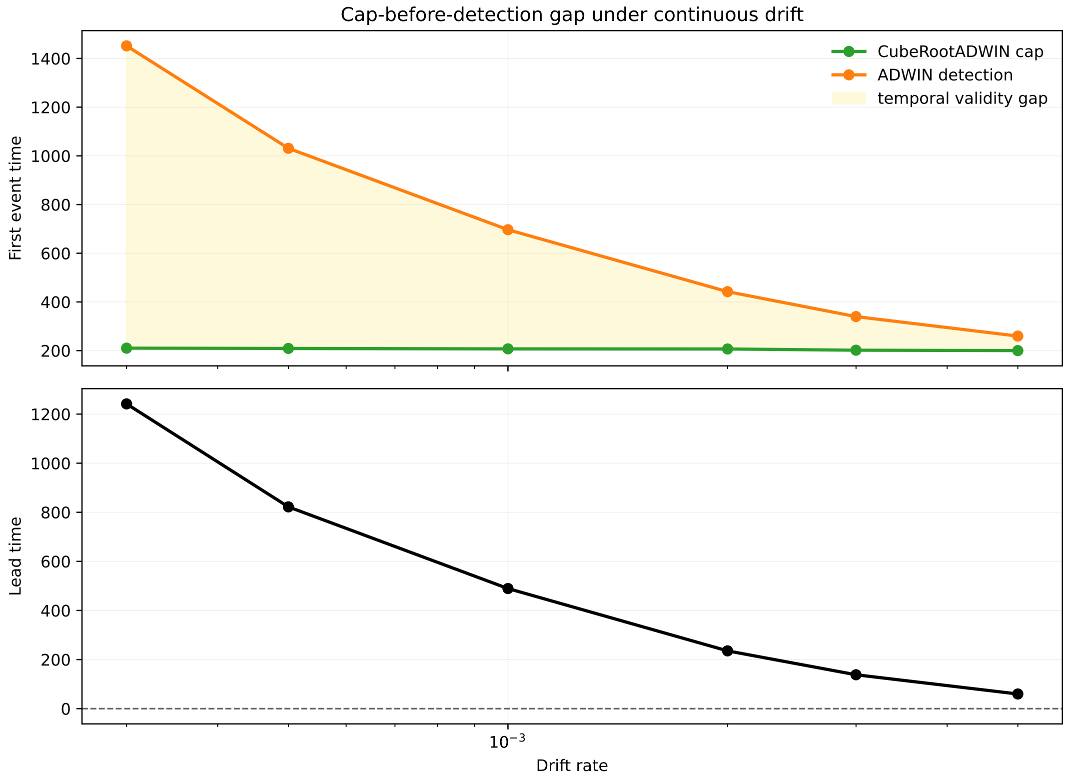

Figure 4: Temporal validity gap across drift rates. The top panel compares the first cube-root cap activation with the first ADWIN detection. The bottom panel shows the resulting lead time. Under slow drift, memory becomes operationally stale long before changepoint evidence is available; under faster drift, the gap narrows but remains positive.

3.5 Piecewise Horizon Recovery

The piecewise-drift benchmark changes the local drift rate in three phases, and the adaptive horizon is compared with the offline oracle fixed window for each phase. The calibration is held fixed across phases; only the stream changes.

The central distinction is between statistical detection and temporal validity. Statistical detection asks whether there is enough evidence to declare a change. Temporal validity asks whether the memory being used is still fresh enough to count as current. The cap-only regime makes that gap visible.

Table 2: Oracle horizon recovery on piecewise drift. The same calibrated cap tracks the oracle scale across regimes: it is close in the fast-drift phase and remains within the right order of magnitude in the smoother phases.

Drift	Oracle window	Cube n^*	Oracle MAE	Cube MAE
0.0005	200	191.2	0.0741	0.0874
0.003	50	58.1	0.1258	0.1381
0.001	100	89.7	0.0876	0.1033

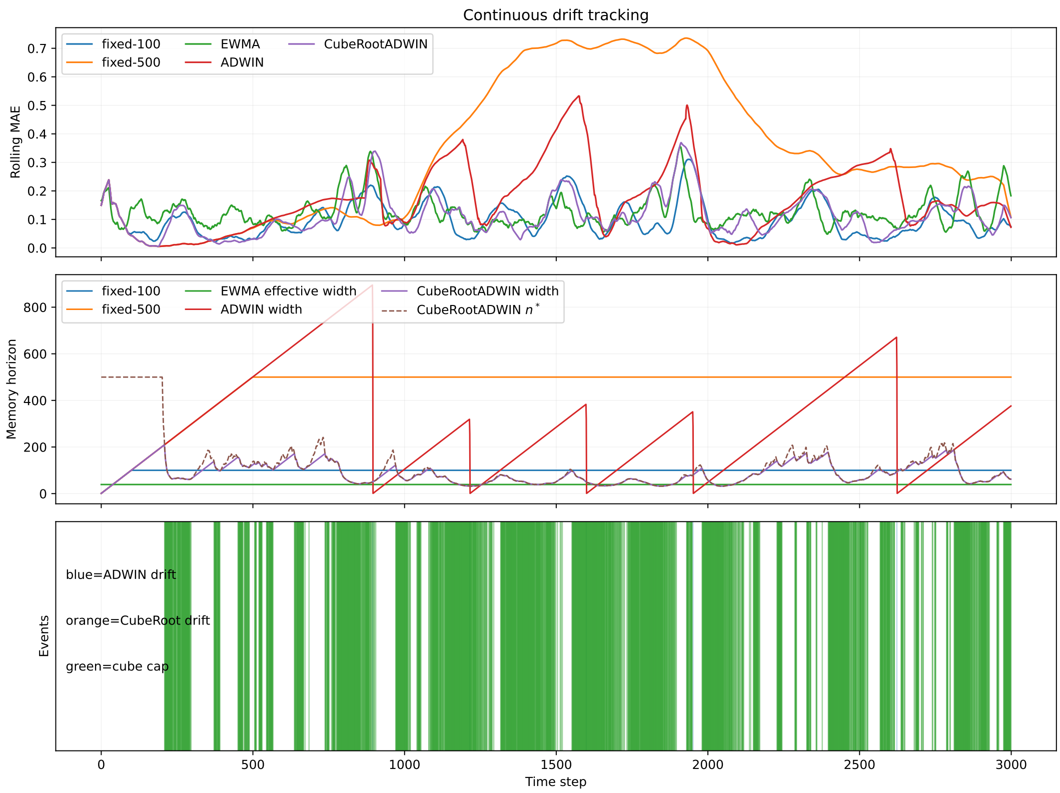

Figure 5: Piecewise-drift recovery. The same cube-root calibration follows the phase-dependent oracle window closely enough to preserve the right operational scale in each regime. The result is not a claim of universal superiority; it is evidence that the law can be turned into a usable horizon regulator.

3.6 What the Figures Show

Taken together, the figures support three claims. First, the cube-root law is a real finite-memory law rather than a fitting artifact. Second, adaptive windows that optimize detection can still accumulate too much stale memory under smooth drift. Third, the cube-root cap gives an explicit operational rule for keeping the horizon of useful memory near the oracle scale.

4 Related Work

Tracking under nonstationarity. Dynamic regret, online least-squares, and adaptive filtering all ask a closely related question: how much past evidence should remain relevant once the target is moving [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. The present paper keeps that question in distributional form and makes the memory horizon itself the main object of analysis.

Transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural because they respect the geometry of the signal space and behave well under transport of mass and support. Recent finite-sample results clarify how the constants depend on geometry and regularization, while leaving the exponent structure intact [Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. Here, the geometry quantifies how much old evidence can be retained before it becomes stale.

Adaptive forgetting and horizon control. Classical smoothers such as EWMA, forgetting-factor RLS, and Kalman filtering all instantiate the same tension between variance reduction and responsiveness. What they often leave implicit is the temporal validity of the retained memory. The cube-root law makes that validity visible by turning the trade-off into an operational horizon selector.

Freshness and timeliness. The Age-of-Information literature gives a useful interpretive bridge: recent information is more timely, and timeliness carries a cost. That viewpoint fits the argument here well, while AoI remains an interpretation rather than a modeling target.

What is different here. The result sits next to drift detection, but its organizing distinction is a different one: statistical evidence of change versus the horizon at which the retained memory remains current.

5 Limitations and Open Directions

Scope of the law. The cube-root law is exact for the finite-memory averaging class analyzed in this paper, and the lower bound is stated for window-restricted estimators on a W_2-Lipschitz drift class. It captures a structural finite-memory phenomenon, though not a universal minimax theorem over all possible online estimators.

Dependence on local drift estimation. The operational regulator depends on a local estimate of drift. The current benchmark suggests that the remaining gap to the oracle comes mostly from drift estimation. Better local geometry estimators may therefore sharpen the horizon recovery without changing the law.

Geometry beyond the current metric. The present formulation uses Wasserstein geometry and a debiased Sinkhorn measurement layer when samples must be compared computationally. Other geometries may admit analogous trade-offs, but the constants and the calibration of useful memory may change.

What the experiments show. The current experiments are not a leaderboard over drift detectors. They show a specific operational distinction: a tracker can still have statistically insufficient changepoint evidence while already being stale in the sense of useful memory.

6 Conclusion

How long can memory remain useful when the target distribution keeps moving? Under the drift class studied here, useful memory has a finite temporal horizon, and that horizon obeys a cube-root law.

The theoretical contribution is a decomposition of tracking error into a variance term and a staleness term, together with a lower bound showing that the resulting finite-memory floor is structural. The operational contribution is a memory regulator: a cube-root cap keeps the effective horizon near the oracle scale across drift regimes, while classical adaptive windows can still accumulate stale memory.

Empirically, the signature regime is 'cap-only': the memory horizon becomes operationally stale before there is enough statistical evidence for a changepoint alarm. That separation between evidence of change and temporal validity of memory is the main conceptual payoff of the paper.

Finite-memory systems are best judged by how long their retained evidence remains current enough to count as useful. The paper contributes a law with a clear design consequence: regulate useful memory and track the age at which evidence stops being current.

The effective-memory-age viewpoint sharpens the rule further: the key question is how old a retained history may become before it stops being operationally current.

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A Code Repository

All implementations are publicly available at:

https://github.com/jovemexausto/coherence-delay-tradeoff

The repository contains:

- `experiments/run.py` — single entry point for all experiment domains;
- `experiments/gaussian/` and `experiments/particle/` — core experiment logic;

- `experiments/bikes/`, `experiments/elec2/`, and `experiments/kuairand/` — domain-specific benchmarks;
- `REPRODUCIBILITY.md` — setup and run instructions.

All experiments use `uv` for dependency management (`uv sync`) and are reproducible with fixed random seeds as specified in each script.